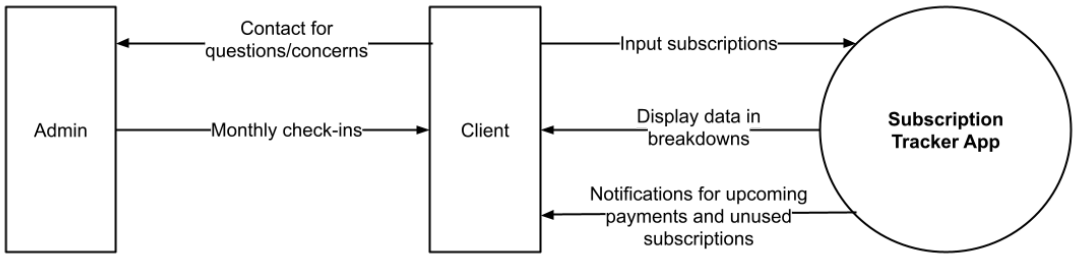
Design Diagrams:

System Context Diagram

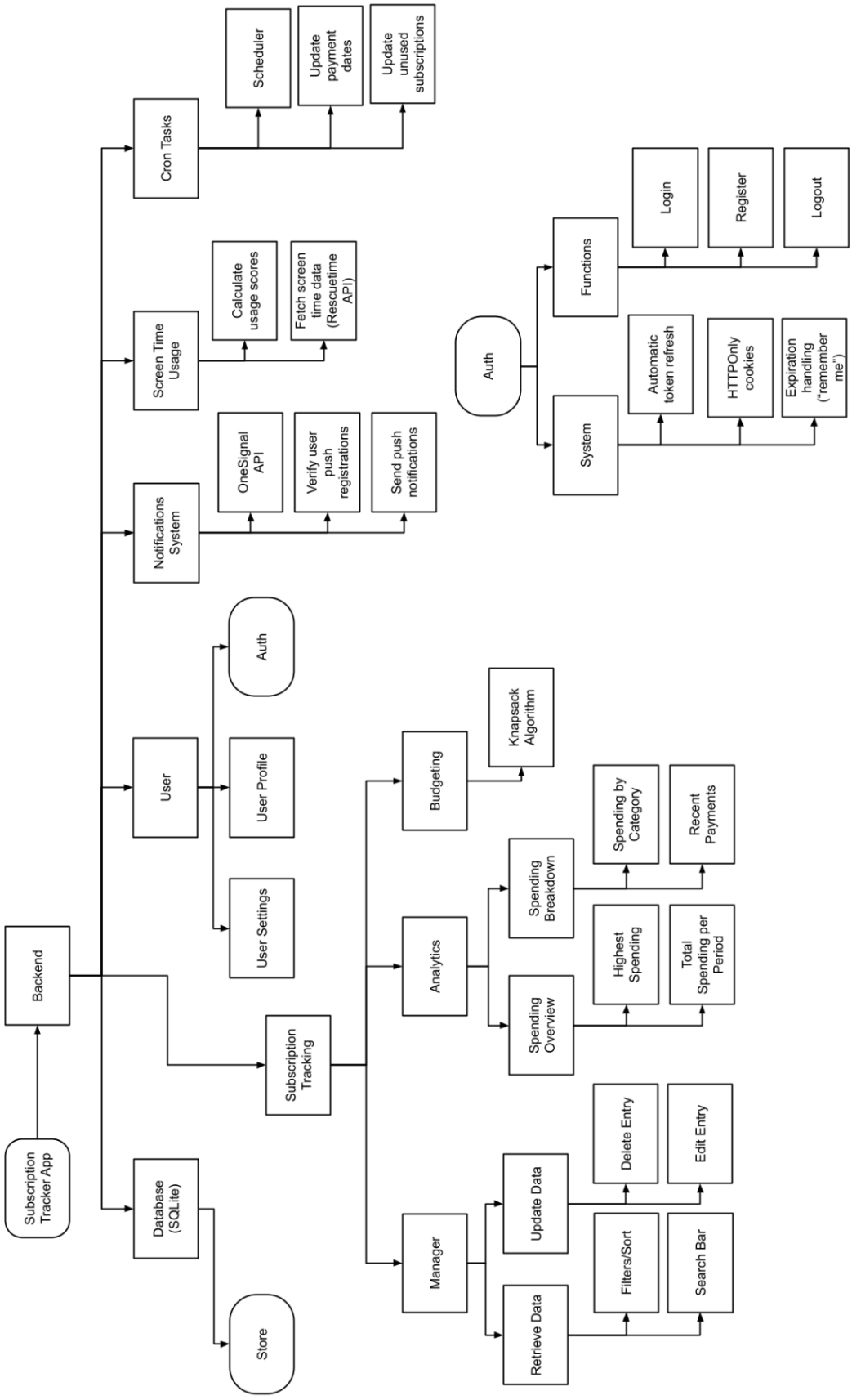


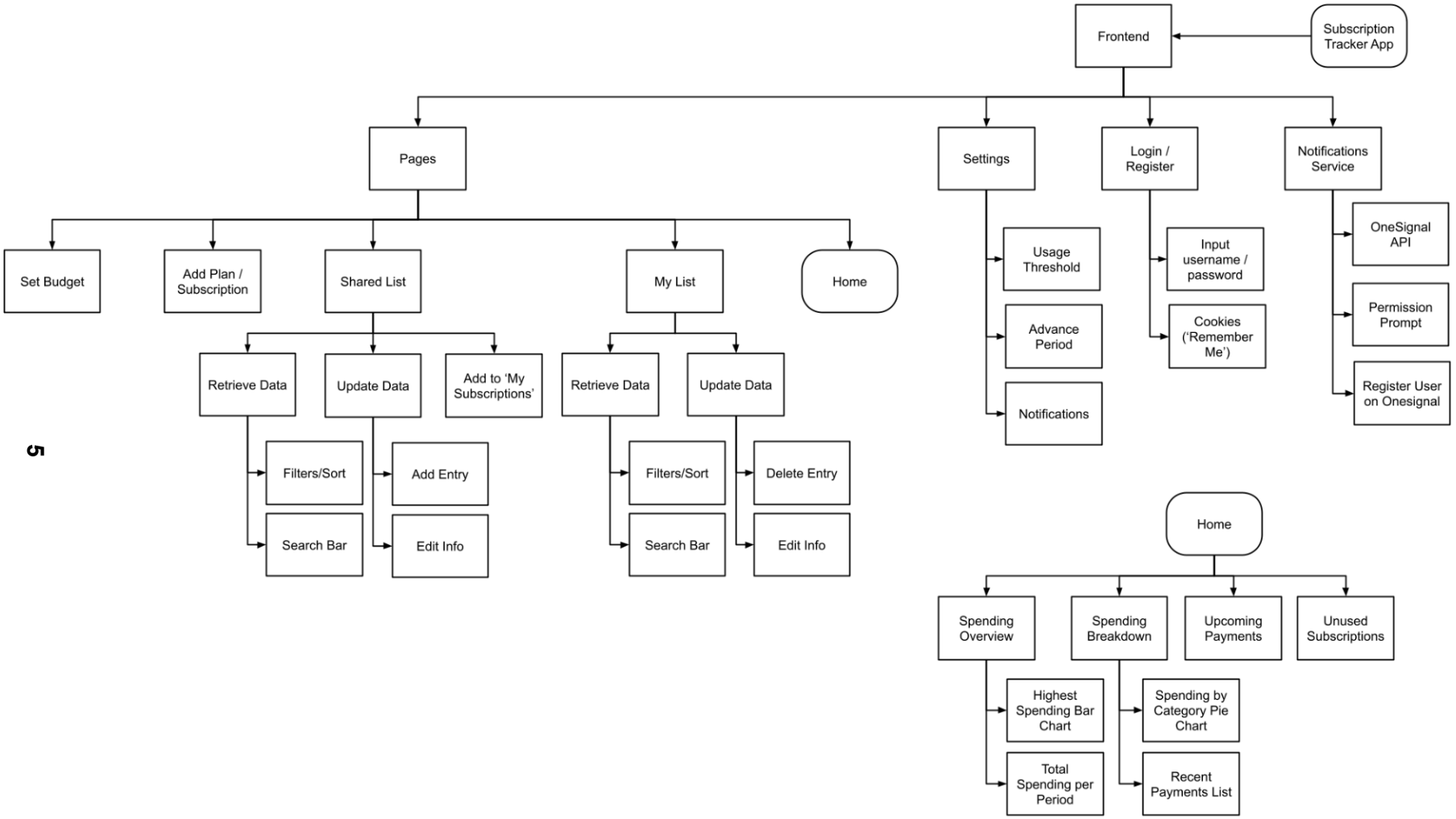
Design Overview

Problem(s)	**Views**	**Control**	**Models**
User tasks / jobs	*Modules (interfaces or classes)*	*Algorithms (methods)*	*Data items (objects, lists, etc)*
Authentication	CustomAuthHandler useUser (frontend composable)	login()	User model userState (frontend store)
		register()	
		logout()	
		refresh()	
Access shared subscriptions list.	SubscriptionsView	get_queryset()	Subscriptions model
Add new subscription item.		create()	
Delete subscription item.		destroy()	
View payment plans	PaymentsView	get_queryset()	Payment plans model
Add payment plans to a personal list.		create()	
Edit payment plan.		update()	
Delete payment plan from list.		destroy()	
Search by name and apply filters to sort by attributes.	ModelManager	search()	Subscriptions model Payment plans model User plans model
		apply_filters()	
		sort_by()	
Calculate usage scores of subscriptions from screen time data.	ScreentimeManager	fetch_data()	User plan table Screentime data table
		calculate_usage_score()	
Receive and set notifications for unused subscriptions and upcoming payments.	NotificationManager	register_onesignal()	User plan table Plan table
		send_push_notification()	

Set budget		set_budget()	Hashmap
Spending per payment period	SpendingCalculator AverageSpendingCalculator StatisticsView	calculate_spending()	User plan table Plan table
		calculate_range_spending()	
		view()	
Category breakdowns by cost and usage.	CategoryBreakdown	view()	Plan table User plan table

Top Down Design




Frontend
Subscription Tracker App
Pages
Settings
Login / Register
Notifications Service
Set Budget
Add Plan / Subscription
Shared List
My List
Home
Usage Threshold
Advance Period
Notifications
Input username / password
Cookies ('Remember Me')
OneSignal API
Permission Prompt
Register User on Onesignal
Retrieve Data
Update Data
Add to 'My Subscriptions'
Retrieve Data
Update Data
Filters/Sort
Search Bar
Add Entry
Edit Info
Filters/Sort
Search Bar
Delete Entry
Edit Info
Home
Spending Overview
Spending Breakdown
Upcoming Payments
Unused Subscriptions
Highest Spending Bar Chart
Total Spending per Period
Spending by Category Pie Chart
Recent Payments List

Data Dictionary

User

Field Name	Data Type	Constraints	Description	Example
id	serial	PRIMARY KEY, NOT NULL	Unique identifier for each registered user	1
name	varchar	LIMIT 64, NOT NULL	Username of registered user	Guest
password_hash	string	LIMIT 128, NOT NULL	Encrypted version of user's password	
remember_me	boolean	DEFAULT FALSE	Opt-in for cookies to keep users logged in across sessions	True
allow_notifications	boolean	DEFAULT FALSE	Opt-in to be notified on upcoming payments and unused subscriptions	True
rescuetime_api_key	varchar	LIMIT 255	Encrypted Rescuetime API key	
advance_period	integer	DEFAULT 3 UNSIGNED	Number of days prior to a payment date to alert the user	3

Subscription (Shared)

Field Name	Data Type	Constraints	Description	Example
id	serial	PRIMARY KEY, NOT NULL	Unique identifier for each subscription	1
name	varchar	NOT NULL	Name of the subscription	RBC
category_id	int	FOREIGN KEY, NOT NULL	Links to Categories table	Finance
icon_url	url		Image url of the subscription	media.rbc.com/logo

Plan (Shared)

Field Name	Data Type	Constraints	Description	Example
id	serial	PRIMARY KEY, NOT NULL	Unique identifier for each payment plan	2
subscription_id	integer	FOREIGN KEY, NOT NULL	Links to Subscriptions table	1
name	varchar	NOT NULL	Name of the plan	Advantage Plan
cost	decimal	NOT NULL, DECIMAL PLACES 2	Cost per payment period	25.20
period	varchar	NOT NULL, CHECK(_ IN (“Monthly”, “Yearly”, “Weekly”))	Payment frequency. Must be in “Monthly”, “Yearly”, or “Weekly”	“Monthly”
free_trial	boolean	DEFAULT FALSE	Plan type is a free trial (if true, cost and period are nullified)	False

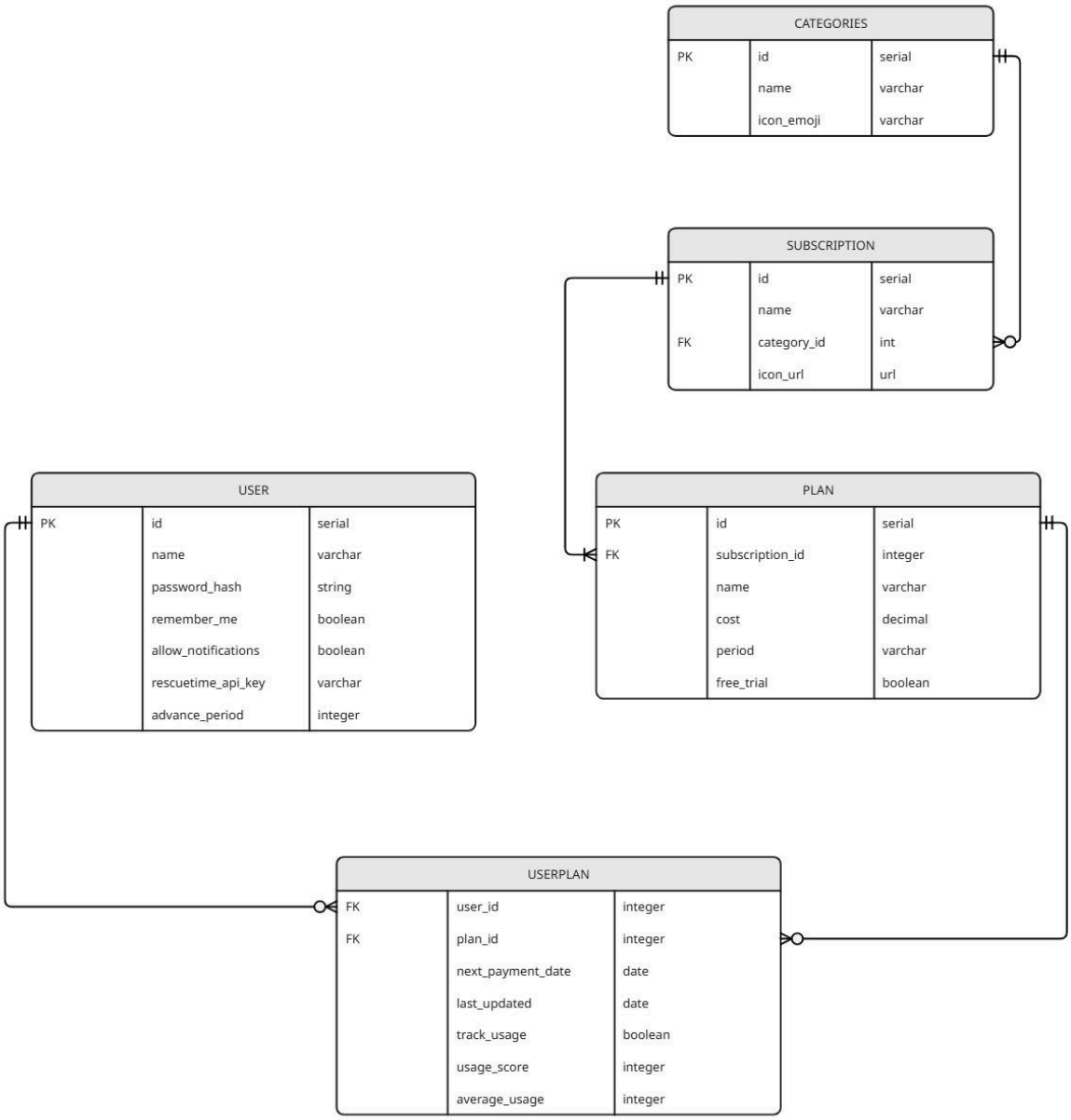
UserPlan

Field Name	Data Type	Constraints	Description	Example
user_id	integer	NOT NULL	Links to User table	3
plan_id	integer	NOT NULL	Links to Plan table	2
next_payment_datae	date	NOT NULL	Next scheduled date for a plan’s payment	2/2/2025
last_updated	date	NOT NULL	Last date payment dates were updated	1/29/2025
track_usage	boolean	DEFAULT FALSE	Opt-in to track the specific subscription’s screen time	False
usage_score	integer	INTEGER, VARCHAR	Grades the usage of a subscription from 1-10	5
average_usage	integer	DEFAULT 0	Average screen time usage of a subscription in seconds per day	3400

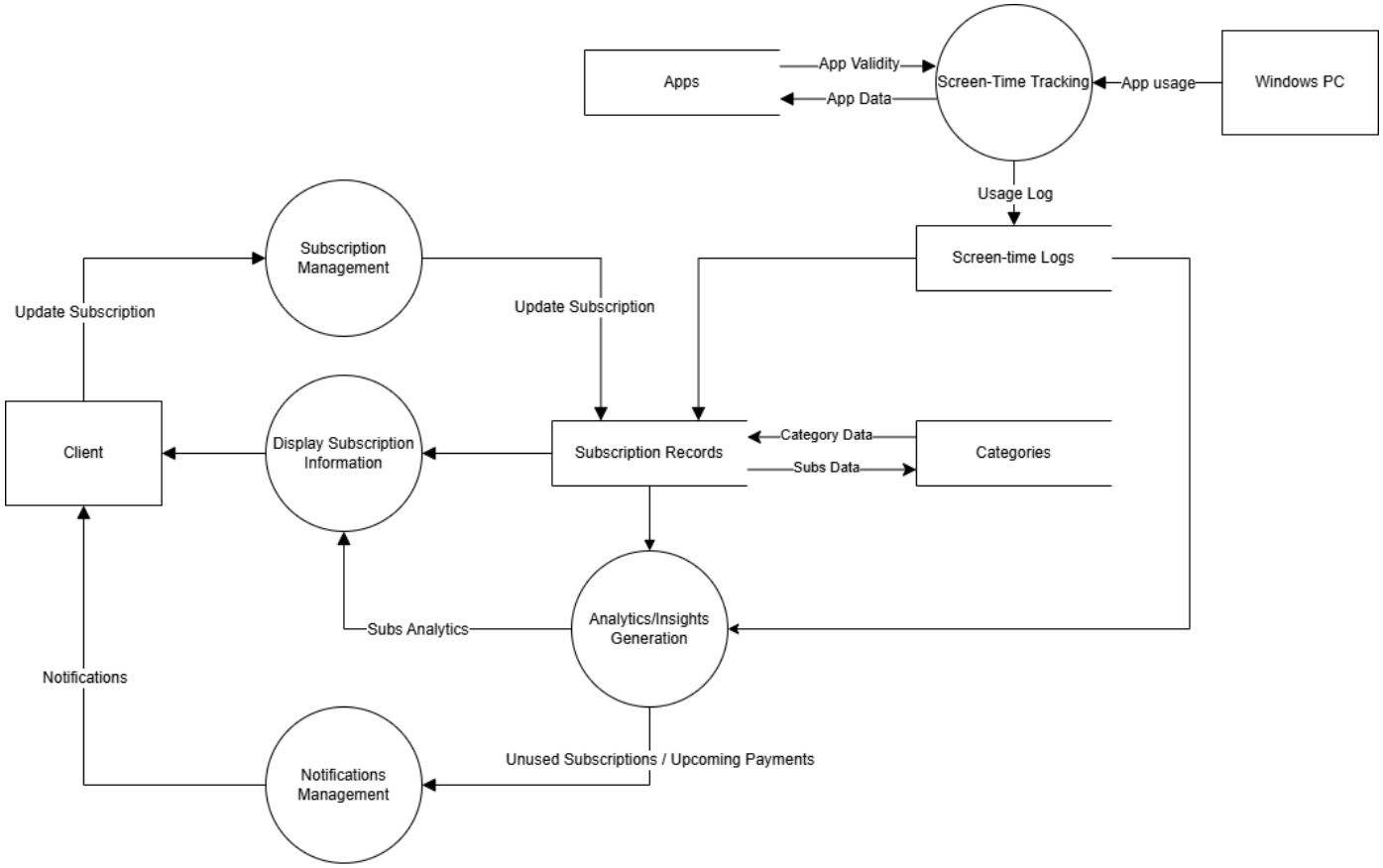
Categories

Field Name	Data Type	Constraints	Description	Example
id	serial	PRIMARY KEY, NOT NULL	Unique identifier for each category	1
name	varchar	NOT NULL	Name of the category	Investing
icon_emoji	varchar	MAX LENGTH 10	Emoji to represent the category	

Entity Relationship Diagram



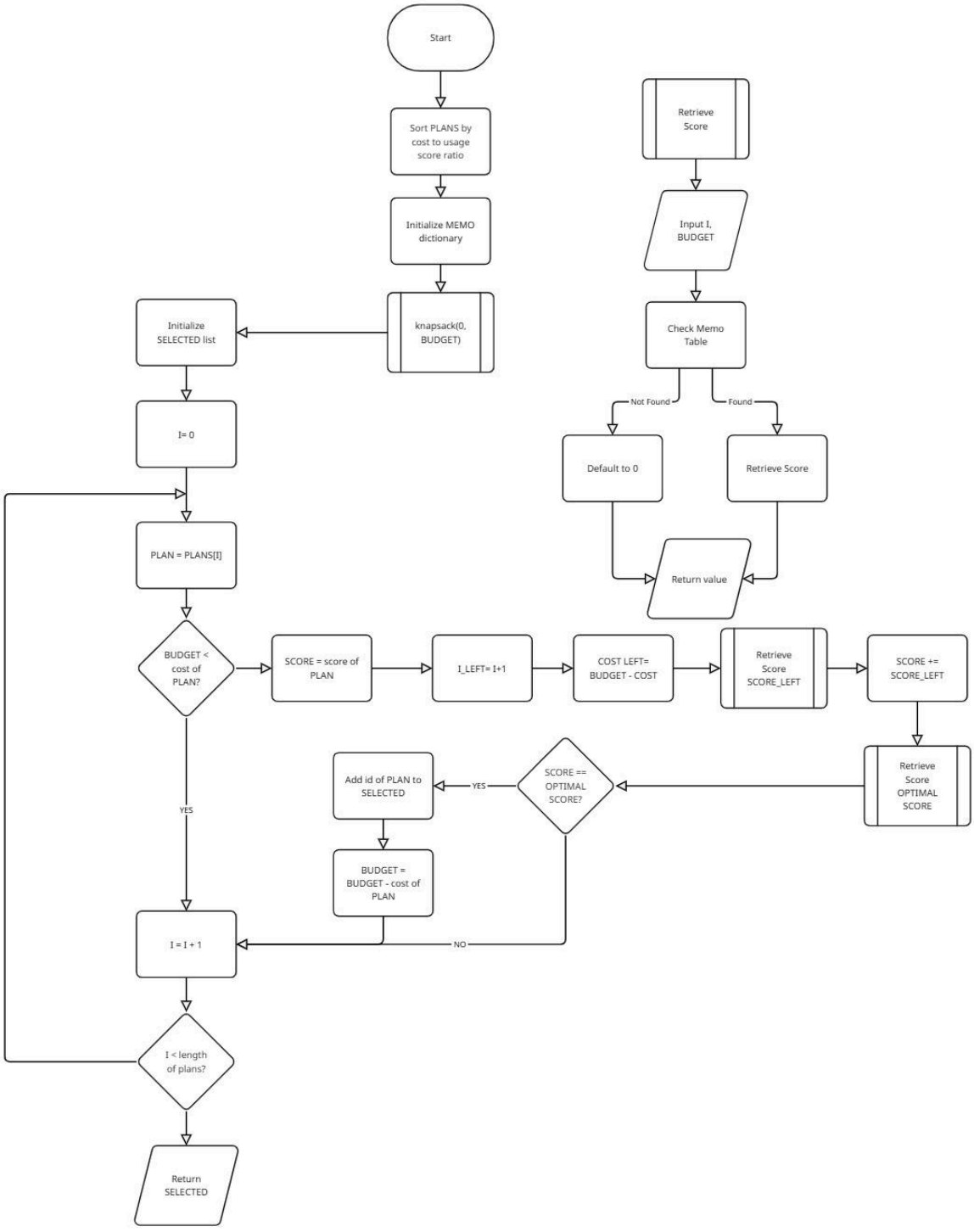
Data Flow Diagram



Important Algorithms

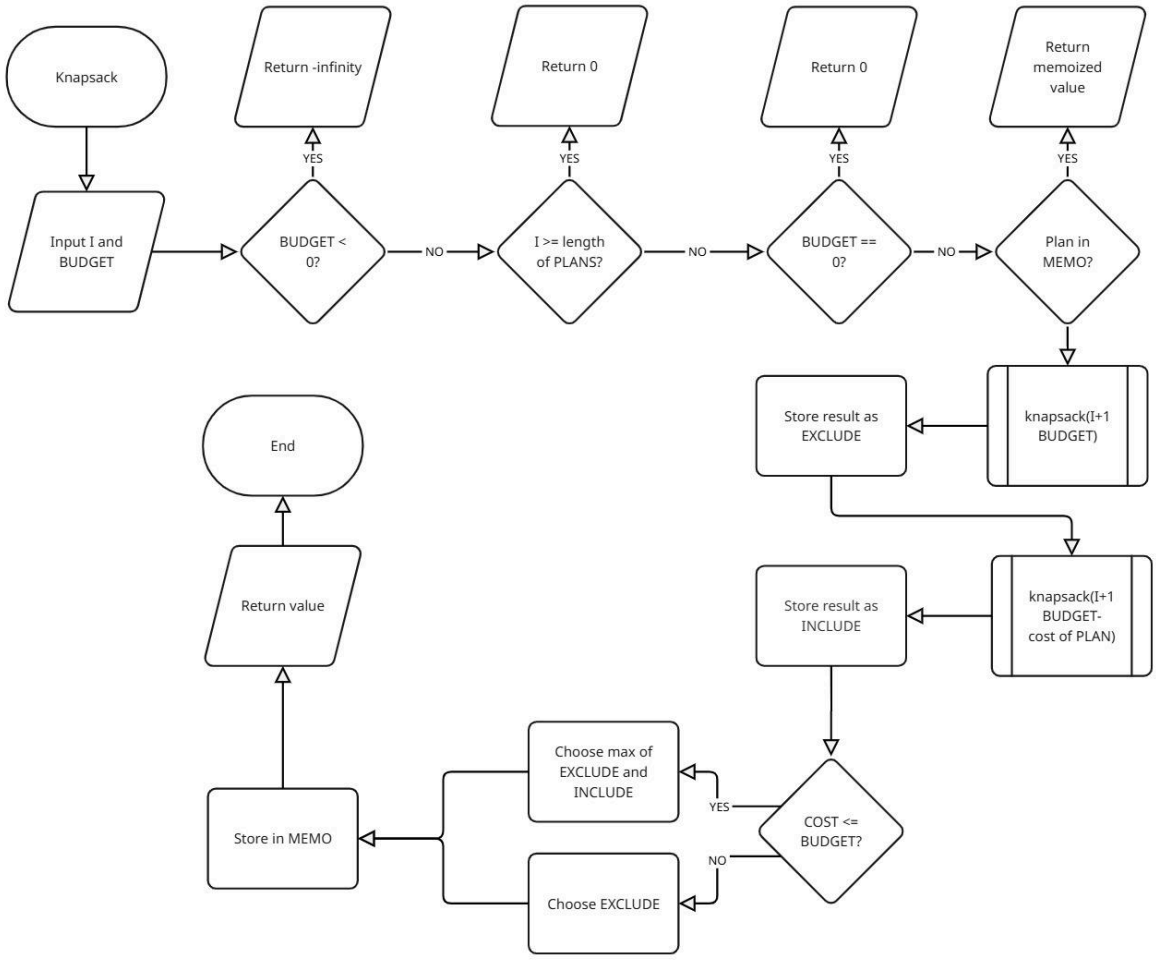
Set Budget

The set budget function takes in a maximum cost (budget) and list of plans, and attempts to trim the number of plans to be within budget while maximizing their value (usage score).
As the algorithm used is specific, it requires some pre and post processing.



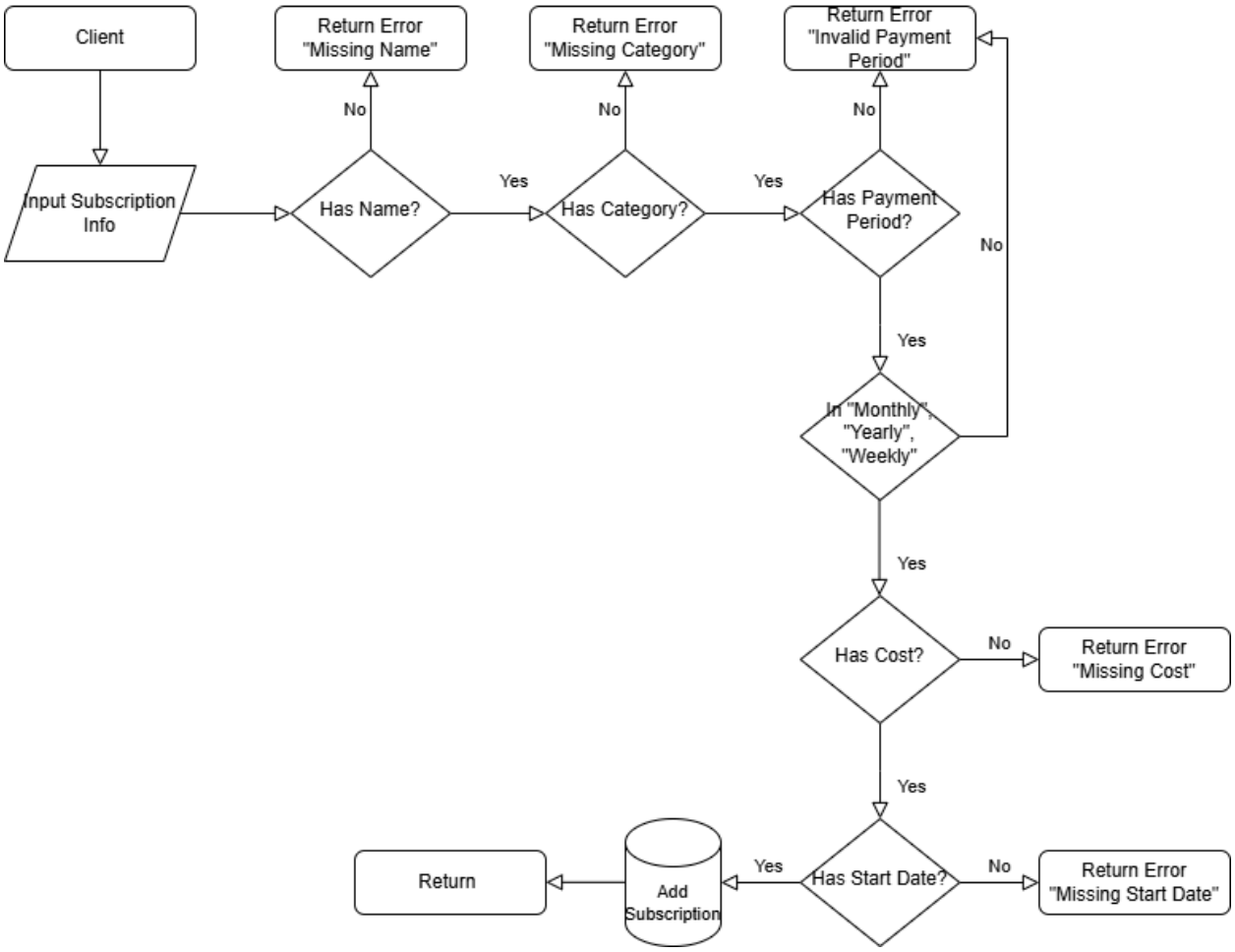
Set Budget (Knapsack)

This is the core algorithm of the set budget function.

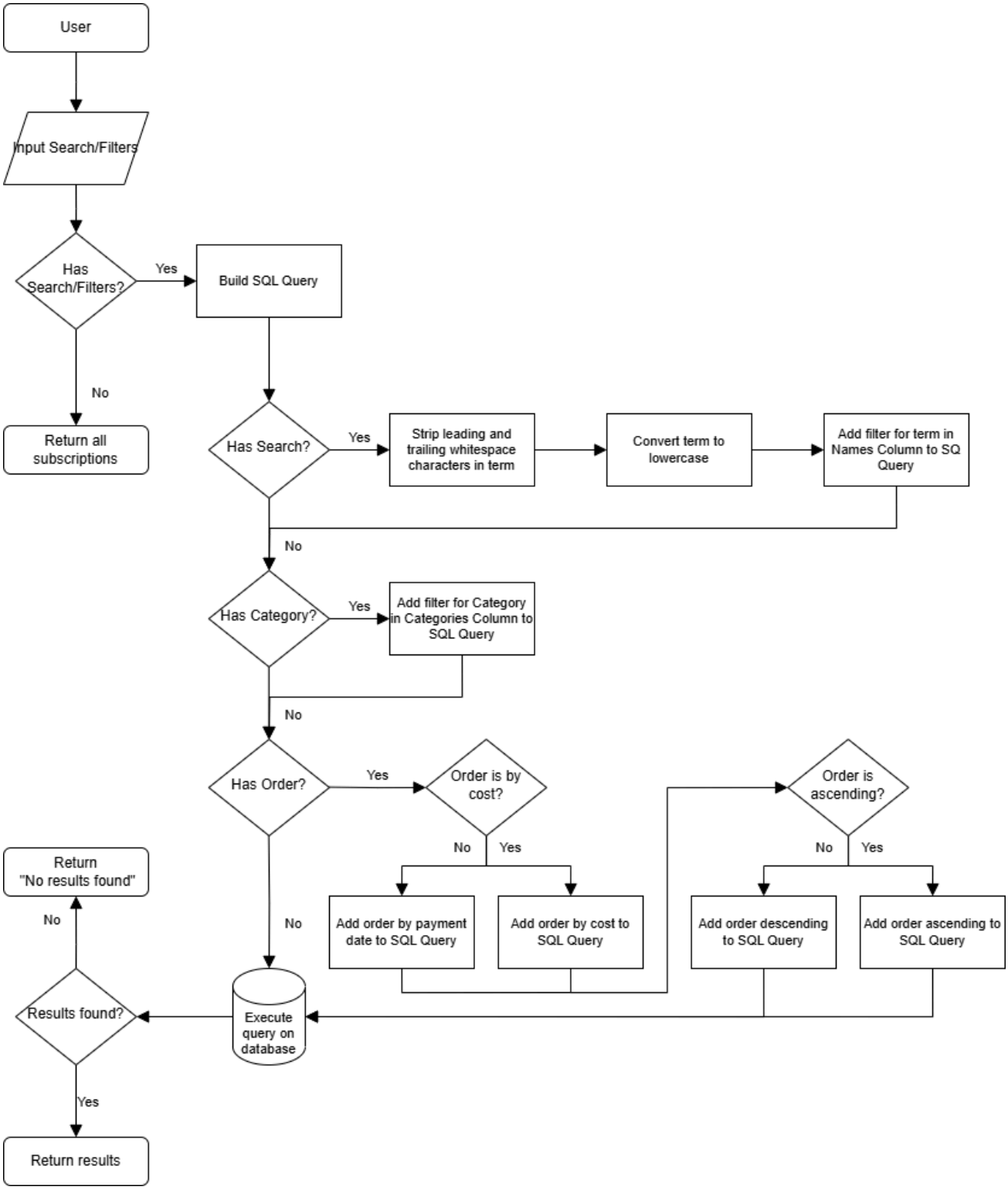


Add New Subscription

When a user adds a new subscription, the program must confirm the inputted information is valid.



Filter and Search Subscriptions

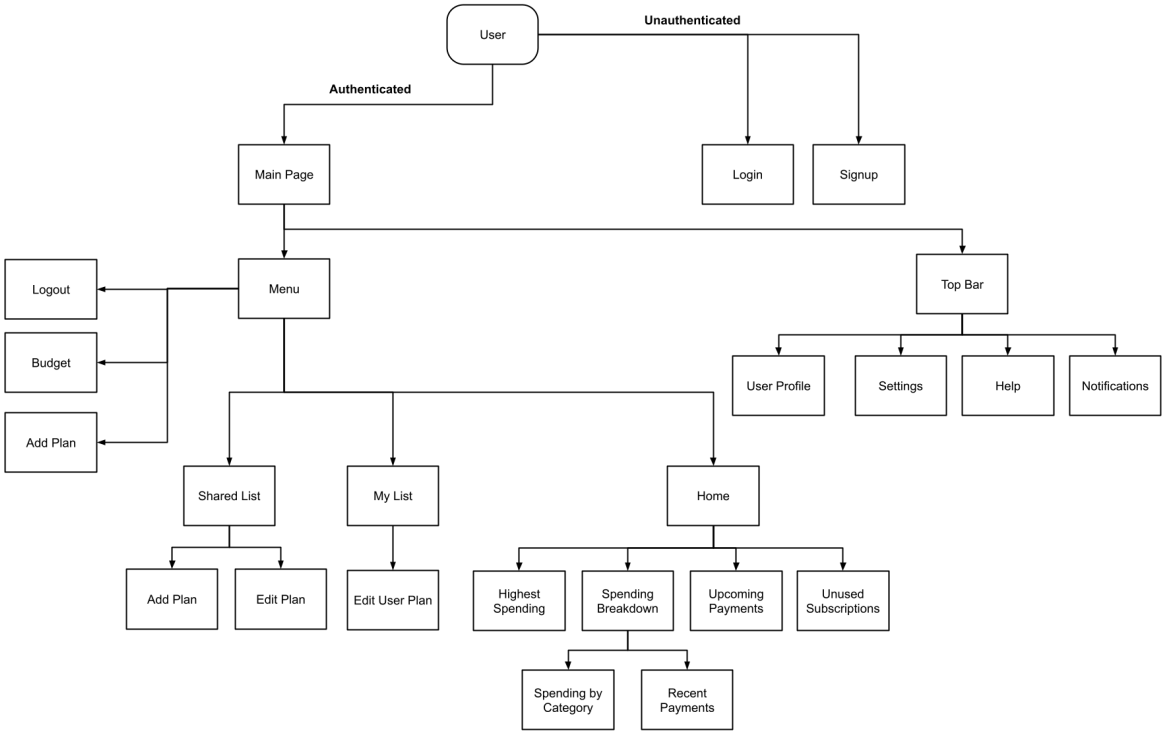


API Routes

Endpoint	HTTP Method	Requires Auth?	Purpose
/api/auth/login	POST	No	Authenticates user and sends cookies with tokens
/api/auth/logout	POST	Yes	Remove authentication cookies from user
/api/auth/register	POST	No	Creates a new user (if they don't exist)
/api/auth/verify	POST	Yes	Verifies user's authentication status, and notifies if expired
/api/subscriptions	GET, PATCH, POST	Yes	Manage and create subscriptions, including nested plan details
/api/plans	GET, PUT, POST	Yes	Access and manage specific plan details
/api/user-plans	GET, PATCH, POST	Yes	Retrieve and manage a user's own plans
/api/categories	GET	Yes	Fetches category details
/api/user/settings	GET, PATCH	Yes	Manage a user's settings (e.g. usage threshold and notifications)
/api/user/profile	GET	Yes	Fetches user name, icon, and other info
/api/analytics/usage-by-category	GET	Yes	Calculates total screen time usage by category
/api/analytics/spending-by-category	GET	Yes	Calculates total spending per category normalized over time periods
/api/analytics/total-spending-per-period	GET	Yes	Calculates total spending over past time period (week, month, quarter, year)
/api/analytics/average-spending-per-period	GET	Yes	Calculates total normalized spending over past time period
/api/cron/payment	POST	No	Sends notifications for upcoming payments and updates the next payment date to be the most recent.

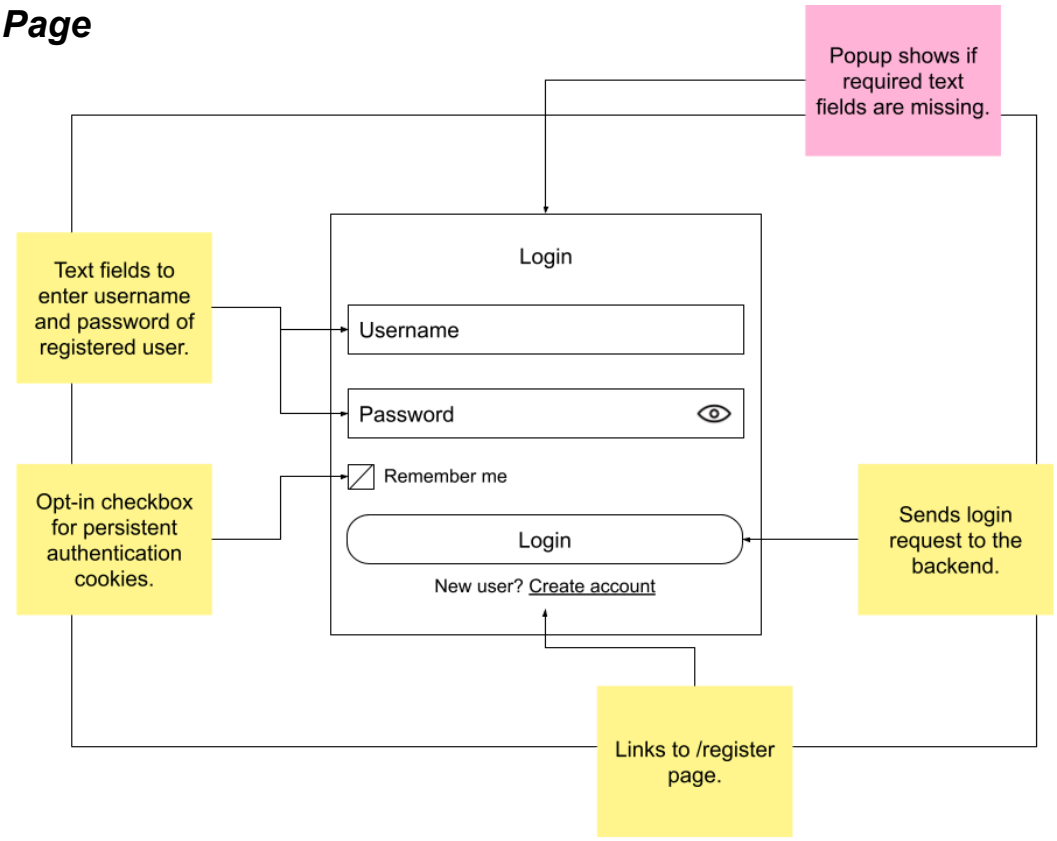
/api/cron/unused	POST	No	Fetches user screen time data, calculates usage scores, and notifies for unused subscriptions

Navigation Diagram

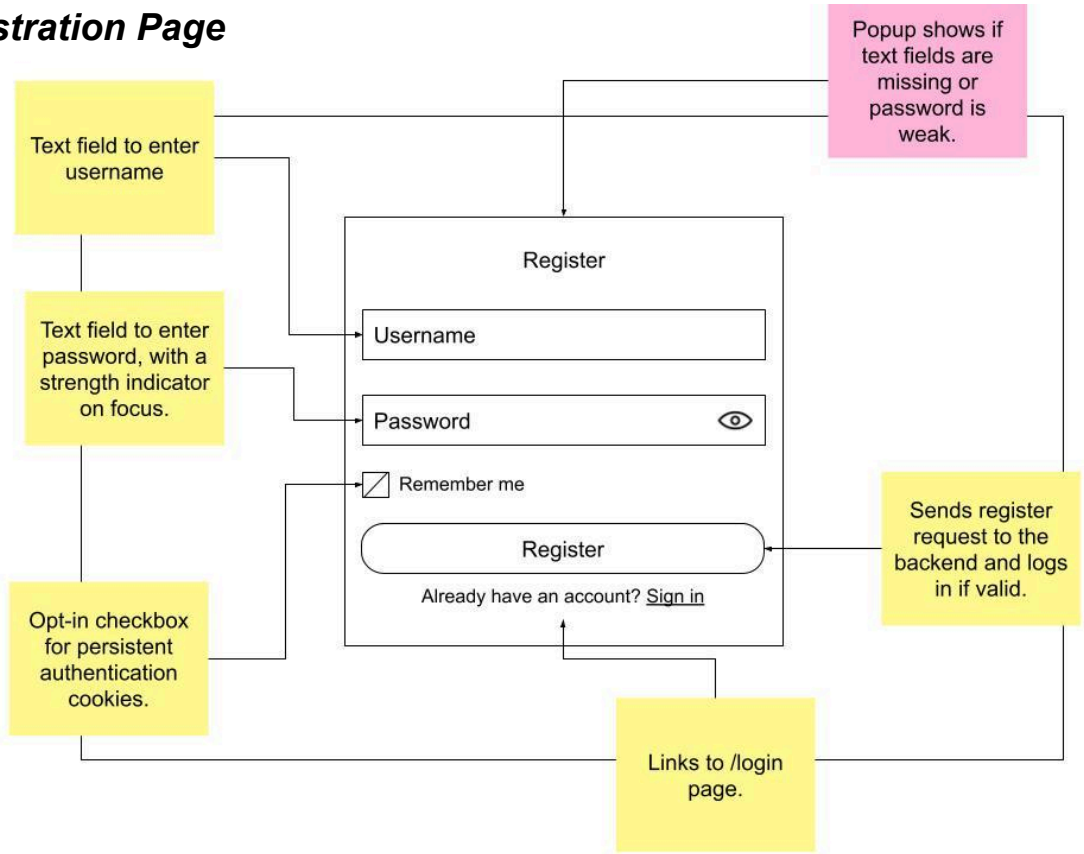


Wireframe

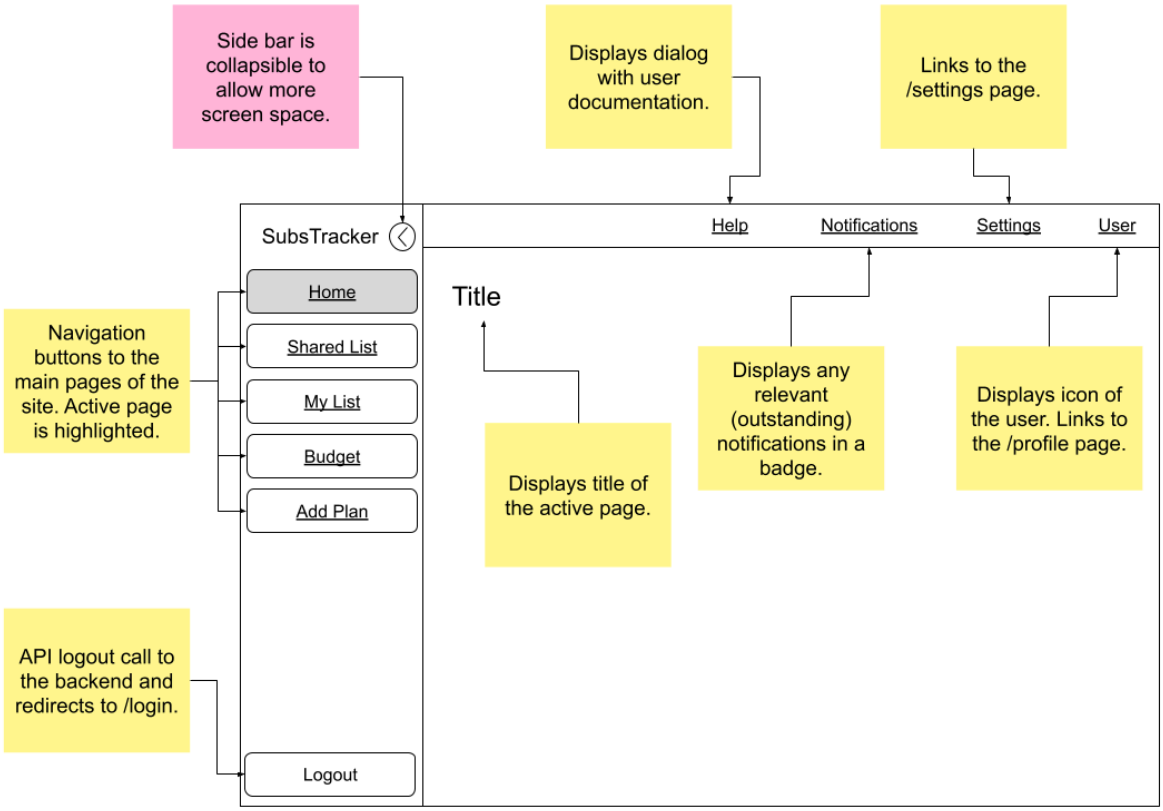
Login Page



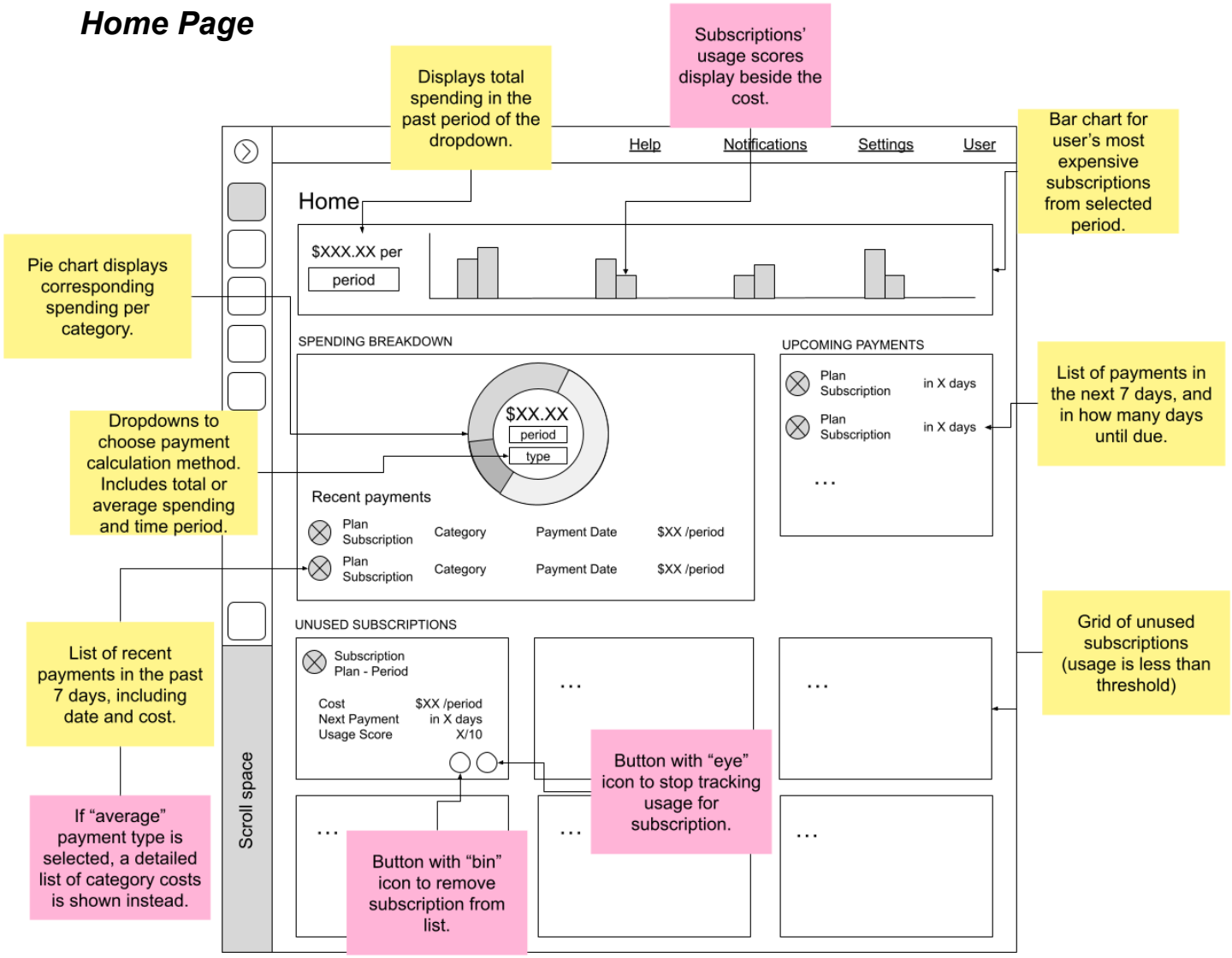
Registration Page



Main Layout

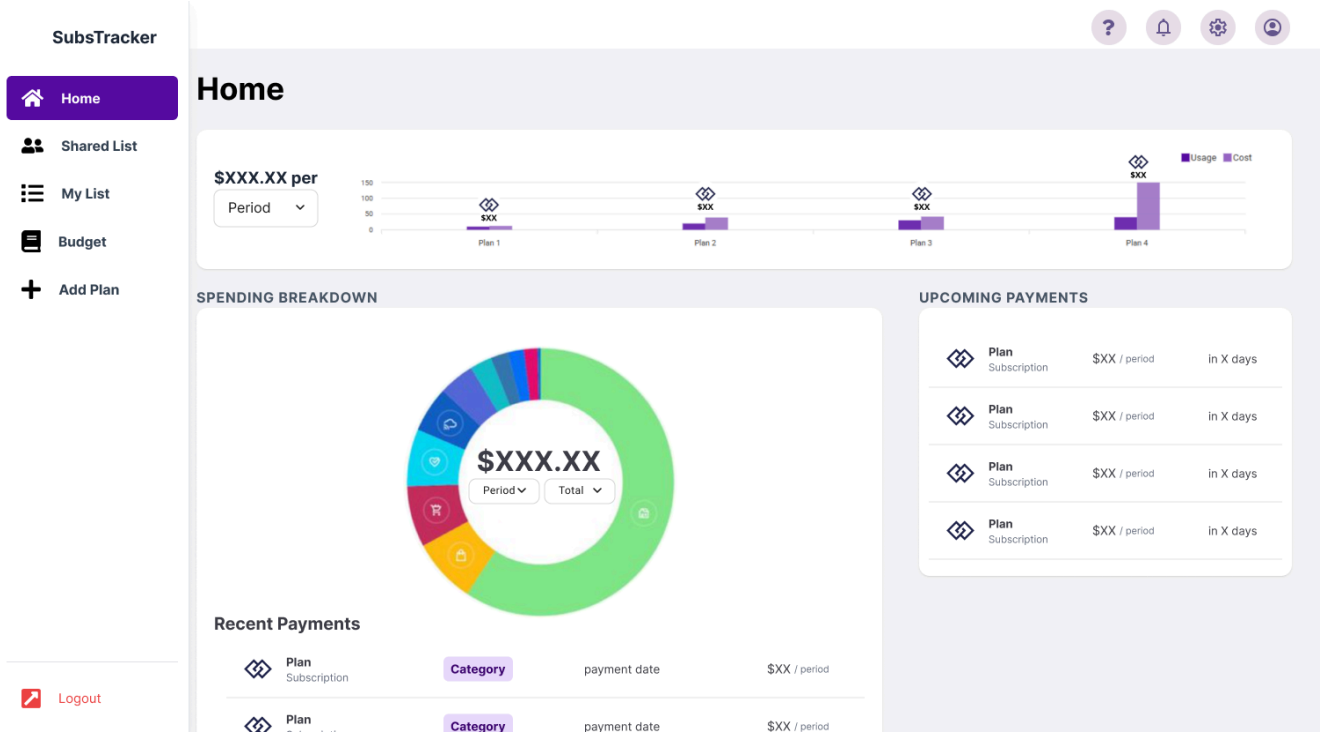

Side bar is collapsible to allow more screen space.
Displays dialog with user documentation.
Links to the /settings page.
SubsTracker
Help
Notifications
Settings
User
Home
Shared List
My List
Budget
Add Plan
Title
Navigation buttons to the main pages of the site. Active page is highlighted.
Displays any relevant (outstanding) notifications in a badge.
Displays icon of the user. Links to the /profile page.
Displays title of the active page.
API logout call to the backend and redirects to /login.
Logout

Home Page

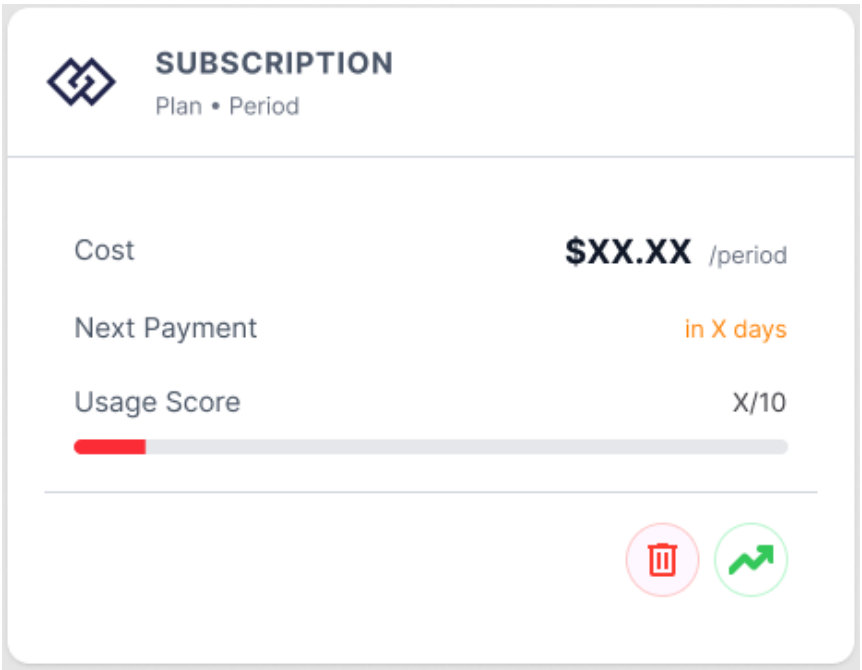

Displays total spending in the past period of the dropdown.
Subscriptions' usage scores display beside the cost.
Bar chart for user's most expensive subscriptions from selected period.
Help
Notifications
Settings
User
Home
$XXX.XX per
period
Pie chart displays corresponding spending per category.
SPENDING BREAKDOWN
$XX.XX
period
type
Dropdowns to choose payment calculation method. Includes total or average spending and time period.
Recent payments
Plan Subscription
Category
Payment Date
$XX /period
Plan Subscription
Category
Payment Date
$XX /period
UPCOMING PAYMENTS
Plan Subscription
in X days
Plan Subscription
in X days
List of payments in the next 7 days, and in how many days until due.
List of recent payments in the past 7 days, including date and cost.
UNUSED SUBSCRIPTIONS
Subscription Plan - Period
Cost
$XX /period
Next Payment
in X days
Usage Score
X/10
Grid of unused subscriptions (usage is less than threshold)
Button with "eye" icon to stop tracking usage for subscription.
Scroll space
If "average" payment type is selected, a detailed list of category costs is shown instead.
Button with "bin" icon to remove subscription from list.

Mockup

Based on the client's feedback, I created a finalized design for the UI in Figma.



I also created the design for the user plan card component, as it would be reused across multiple pages.



Test Plan

Success Criteria	Feature	Method	Expected Results
Secure login and register, and opt-in to stay logged across sessions.	Registration	Open Register page. Enter valid username and password.	User account is created and redirected to the home page.
		Enter blank or missing fields.	‘Missing Field’ messages will display under blank fields.
	Login	Open Login page. Enter valid username and password.	User is logged in and redirected to the home page.
		Enter invalid username or password.	“Incorrect Credentials” error message pops up.
		Enter blank or missing fields.	‘Missing Field’ messages will display under blank fields.
	‘Remember me’	Login with ‘remember me’ checkbox ticked. Close the browser and reopen.	User remains logged in, and will be directed to the home page as usual.
All users can access, edit, and add to a shared subscription list.	Create new subscriptions	Open the Shared List page. Click the ‘Add’ button. Select category and payment period from the respective dropdown menus. Fill in the name, date, and cost fields with valid information. Press confirm.	‘Add Subscription’ page opens. Receive success message. Inputted subscription data will display on the list.
		Have inputted at least 20 subscriptions in the list with valid data.	Display all subscriptions in the list with accurate information.
		Leave all fields blank and press confirm.	Subscription page will not close. ‘Missing Field’ messages will display under the blank fields.
		Enter a negative number or characters in the cost field.	‘Invalid Cost’ will display under the cost field.
		Enter an invalid date or non-numeric characters in the date field.	‘Invalid Date’ will display under the date field.

	Edit subscription	Click the ‘Edit’ button on a subscription and enter valid inputs.	Changes are displayed on the subscription list.
		Input invalid information into fields (same as ‘Add’).	‘Invalid’ message will display under the invalid field and will not save.
	Delete subscription	Click the ‘Delete’ button on a subscription. Press confirm.	Subscription no longer appears in the list.
		‘Delete’ and press cancel.	Subscription stays intact.
Users can add payment plans to a personal list. Here, payment dates and screen time are tracked and managed.	Add to personal list	Open the Shared List page. Click the ‘Add’ button beside a payment plan. Select a valid payment date and press add. Open My List page.	The payment plan will display on the list with valid payment date and subscription information.
	Update payment dates	Add a plan with a payment date for today, and period of week. Open My List. Open Upcoming Payments in the Home page. Wait until the next day. Open My List. Open Recent Payments in the Home page.	The payment plan will display on the list with the valid payment date. The payment plan will display in the list as “in 0 days.” The payment plan’s payment date will be updated to next week’s date. The payment plan will display as “1 day ago.”
Screen time tracking integration for installed services. Unused plans will be flagged based on these metrics.	Track usage	Use an app linked to a subscription for an amount of time.	Subscription will update with the appropriate time in the usage column of the subscriptions list.
	Flag unused plans	Add a plan with no screen time usage. Scroll to Unused section of the Home page.	Subscription is flagged as unused and displays in the list.
	View flagged plans	Open the Home page. Scroll to Unused section.	Displays list of unused subscriptions with usage scores under the threshold.
Users can set up notifications for due payments	Notify unused subscriptions	Open Settings. Set unused usage threshold.	Receive notifications for subscriptions below usage threshold.

and unused plans with user defined parameters.	Notify upcoming payments	Open Settings. Set an advance period.	Receive notifications for payment with correct advance.
Users can search for subscriptions or plans by name and apply filters and sorts.	Search subscriptions by name	Type a valid search term into the search bar.	Only subscriptions with names containing the term will display on the list.
		Input a term with special or control characters.	Only valid and sanitized characters in the term will be used to filter the list.
		Input only whitespace characters.	Subscription list will ignore the search term and continue to apply other filters.
		Enter a long string (50+ characters).	Search term will be limited to the first 50 characters, and displayed accordingly in results.
	Filter through subscriptions by category, payment period, cost, and time usage.	Select a category from the category dropdown filter. Press ‘Apply’.	Only subscriptions with the selected category are returned.
		Select a payment period from the payment filter. Press ‘Apply’.	Only subscriptions with the selected period are returned.
		Drag the cost range slider to be within a selected value. Press ‘Apply.	List only displays subscriptions within the specified cost range.
		Click the usage or cost buttons to filter by their highest/lowest.	Results are ordered by either highest/lowest in the cost or usage categories.
Users can set a budget to specific categories, and receive advice to stay within budget.	Set budget	Enter a maximum budget cost in the number field. Select a category from the dropdown. Press ‘Set Budget’.	Budget should be successfully set and displayed for the selected category.
		Input a large (>$9999) or negative budget.	Invalid input error displays.
		Enter blank fields for budget cost.	Missing field error displays.

Users can view detailed analytics.	Pie chart for overall spending and spending by category.	Open the Dashboard page. Click on ‘Overview’ heading.	Display a pie chart with accurate overall spending and percentage total spending by category.
	List of categories with categorical metrics.	Open the Dashboard page. Click on ‘Categories’ heading.	Display a list of all categories with total spending and usage per time period.